SUMIT THAKUR BARAHI

Mechanical Engineer

Kathmandu, Nepal

Research Interests

A Mechanical Engineering student with interests in control systems, robotics, and applied AI/ML. Focus on implementing theoretical concepts—particularly control theory and reinforcement learning—into autonomous systems. Also passionate about 3D CAD design, simulation, and embedded systems development for practical engineering applications.

Education

Bachelor of Engineering (B.E.) in Mechanical Engineering 2021–2025

Tribhuvan University (TU), Institute of Engineering (IOE)

Pulchowk Campus, Lalitpur, Nepal

Relevant Coursework: Control Systems, Advanced Mechanical Design, Mechanical Design and Simulation, Operational Research, Finite Element Method, Numerical Methods, Probability and Statistics, Mechanics of Solids, Fluid Mechanics, Heat Transfer, Manufacturing and Production Processes, Instrumentation and Measurement

Research Thesis

Bench-top Testing and Validation of Ankle Prosthetics 2024

Senior Design Project — Advisor: Dr. Nawaraj Bhattarai

- Conducted gait analysis and designed the prosthetic prototype in SolidWorks.
- Implemented PID and Hybrid Zero Dynamics (HZD) control.
- Developed cascaded position-velocity control achieving $< 5\%$ tracking error.
- Experimentally validated natural gait tracking.

Research and Professional Experience

Undergraduate Researcher, Robotics Laboratory — IOE Pulchowk Campus Sept 2021–2025

SolidWorks · Catia . ANSYS · Python · Control Systems

- **ABU Robocon 2023 - Nepal Representative:** Lead Mechanical Design Engineer; developed two Semi-autonomous robots capable of targeting and shooting rings onto poles of 3.5m with 80% accuracy using dual silicone molded roller launching mechanism to project the rings ; advanced to quarterfinals among 13+ countries.
- **ABU Robocon 2022 - Nepal Representative:** Assisting Mechanical Design and Fabrication Engineer; Developed two semi-autonomous robots, R1 and R2, for a Lagori-based task: R1 utilized a dual-roller launching mechanism to project balls for target disruption, while R2 employed a scissor-

based gripping mechanism with chain-driven lifting to collect and stack Lagori; advanced to Semi-finals among 12+ countries.

- **Mechanical Design & Fabrication:** Designed ring-throwing mechanisms, roller systems, and four-bar linkages in SolidWorks; Performed structural FEA using ANSYS; fabricated using lathe, milling, drilling, 3D printing, and pneumatics.
- **International Recognition & Award:** 2nd Runner Up & Received Nagase Award (ABU Robocon 2022) for Brilliant Performance, Received Mabuchi Motor Award (ABU Robocon 2023) for best compact design and functionality.
- **Mentorship & Leadership:** Mentor in ABU Robocon 2024 team (Vietnam), Tutor, SolidWorks Fellowship; Mentor, Hardware Fellowship — guided 500+ students.

Product Design Engineering Intern, KIAS-NAAMII 2023–2025

Solidworks · Arduino · KiCad · C/C++ . 3D printing

- **Mobile Microscope Model:** Designed and fabricated a mobile microscope for the final product design.
- **Automation:** Designed and implemented an automated actuation system using a stepper motor driven by an ATmega microcontroller, enabling precise and repeatable mechanical control in mobile microscope.
- **AI Implementation:** Working on Integration of an AI-based image analysis pipeline to assist automated diagnostics, enabling accurate interpretation and reducing dependence on highly skilled operators.

Research and Development Engineering Intern, Orion Space Nepal 2024–3 months

Solidworks · Laser Cutting · Arduino Nano

- Designed, fabricated, and assembled a robotic arm system, including 3D modeling in SolidWorks and preparation of laser-cut structural components.
- Developed the robotic control architecture by designing a custom PCB in KiCad and integrating an Arduino Nano-based controller for coordinated multi-servo actuation, signal interfacing, and precise motion control.

Applied Machine Learning Intern, ANK (Accelerated Komputing) Sept 2024-Nov 2024

Python · CadQuery · LLM Agents · SolidWorks API . Generative Design . Automation

- Developed LLM-driven agents for automated parametric 3D model generation using CadQuery, translating structured natural language inputs into executable CAD operations (achieved 50–70% accuracy).
- Automated SolidWorks modeling using API-driven macros, reducing manual effort in feature creation and parameter updates.
- Built and optimized CAD automation pipelines, improving design iteration speed, geometric consistency, and reproducibility (accuracy increased to 80–90%).
- Debugged and validated parametric modeling logic to ensure constraint integrity and robust feature sequencing.

AI Microdegree, Fusemachine AI) 2025

Deep Learning · Computer Vision · NLP · Reinforcement Learning . ROS . Python

- Completed a six-month intensive program in applied and theoretical Artificial Intelligence.
- Capstone Project: Developed a mobile robot navigation system using Deep Reinforcement Learning (Soft Actor-Critic) integrated with ROS and Gazebo simulation.
- Designed and implemented end-to-end ML pipelines, including environment setup in Gazebo, ROS-based sensor/actuator interfacing, model training, evaluation, and deployment.
- Integrated RL policies with ROS nodes for real-time decision-making, enabling autonomous navigation in simulated environments.

R&D Design and Product Engineer, Yatri Motorcycles) 2025

Catia · Surface Design · Product Development · GD&T . Automotive Engineering

- Designed structural parts, sub-assemblies, and surface components for electric two-wheelers.
- Model and validate components in for manufacturability.
- Assisted in design optimization, tolerance analysis, and documentation.

Coding & Programming Course, Samsung Innovation Campus (SIC) 2023–2024

Python · Data Structures · Algorithms · Problem Solving · Programming Fundamentals

- Successfully completed a four-month intensive training program in coding and programming.
- Developed strong foundations in Python programming, data structures, and algorithmic problem-solving.
- Built practical projects applying structured programming, logical thinking, and debugging techniques.

Technical Skills

Programming: Python, C/C++, MATLAB
Machine Learning: Scikit-Learn, Pandas, NumPy, Matplotlib,
CAD & Simulation: SolidWorks, Catia, Inventor, Alias, Fusion360, ANSYS, FEA, Topology Optimization
Robotics & Control: ROS, PID, MPC, LQR, Hybrid Zero Dynamics, Digital Twins
Embedded Systems: Arduino, Sensor Integration, Motor Control
Tools: Linux, LaTeX, Jupyter Notebook, VS Code, Git, ROS
Manufacturing: 3D Printing, CNC Machining, Pneumatic Systems, Welding

Honors and Awards

2nd Runner-up, Nagase Award, ABU Robocon 2022 2022
Mabuchi Motor Award, ABU Robocon 2023 2023
Runner-up, 3D Design Hackathon (MechTrix) 2023

Teaching and Leadership

U-ThinkCrazy (Part-Time Mathematics Teacher) 2022

- Taught limits and derivative, integration, and its application

Hardware Fellowship Program (Volunteer mentor) 2024

- Mentored control systems to 200+ students.
- Help in developing a curriculum linking mathematics to robotics applications.
- Mentored embedded systems projects.

ABU Robocon Team Mentor 2024

- Mentored team for international competition (Vietnam).
- Guided CAD, control systems, and fabrication.
- Transferred knowledge of sensor fusion and system integration.

Solidworks Fellowship Program (Instructor, Volunteer)

- Taught solidworks to 60+ students.
- Developed a curriculum for basic solidworks tutorial practice.

References

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